UNIVERSITY OF RWANDA

College of Science and Technology

School of ICT | Department of Computer Engineering

Mobile Application System and Design

Lab 6

Push Notifications with Firebase Cloud Messaging (FCM)

Course	Year 3 CSE – Mobile Application System and Design
Date	08 April 2026
Platform	Flutter + Firebase Cloud Messaging
Names	MBABAZI Patrick Straton
Device	Android Emulator

Section 1: Application Screenshots

1.1 Firebase CLI – Google Account Permission

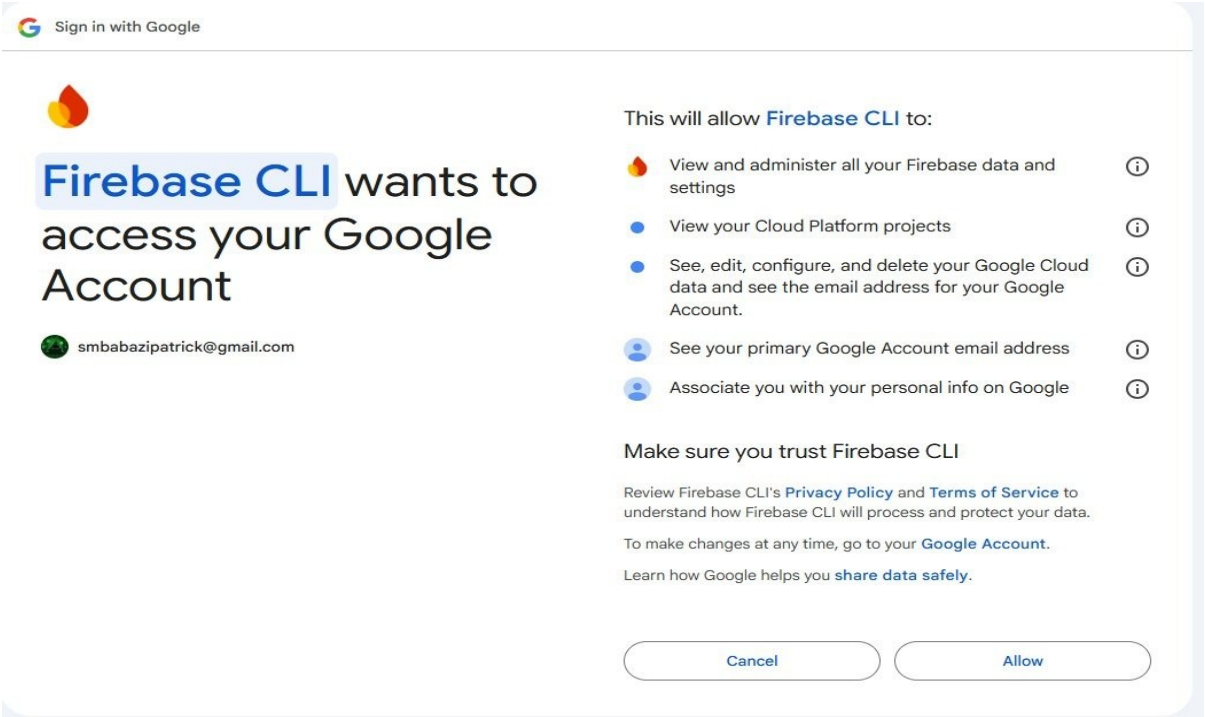


Figure 1 – Firebase CLI requesting Google account permission

1.2 Firebase CLI – Login Successful

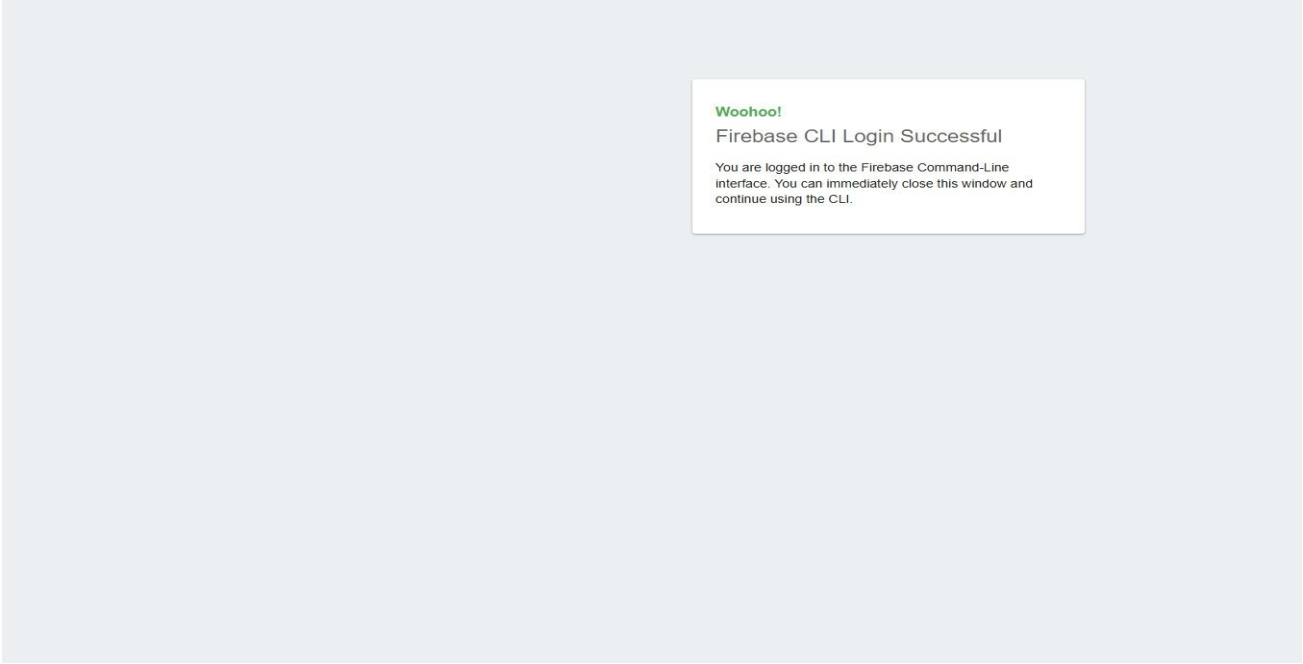


Figure 2 – Firebase CLI login successful confirmation

1.3 App Welcome Screen

The initial screen of the application on first launch.

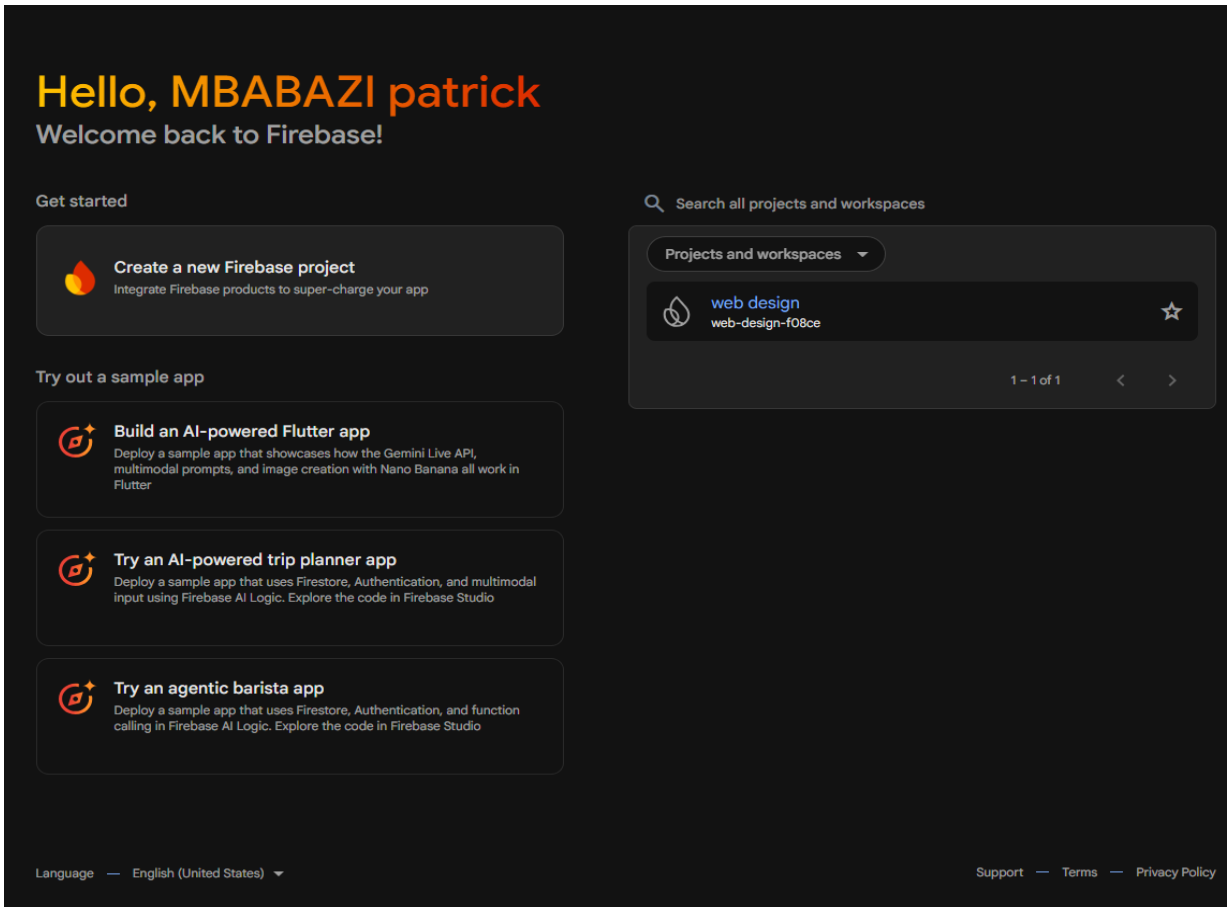


Figure 3 – FCM App welcome screen on launch

1.4 Notifications Enabling on Emulator

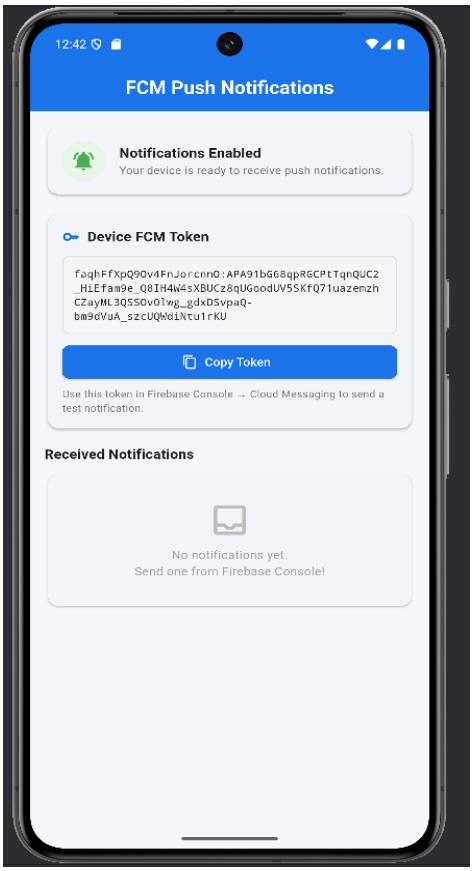


Figure 4 – Notification permission granted, status shows Enabled

1.5 Firebase Console – Cloud Messaging Dashboard

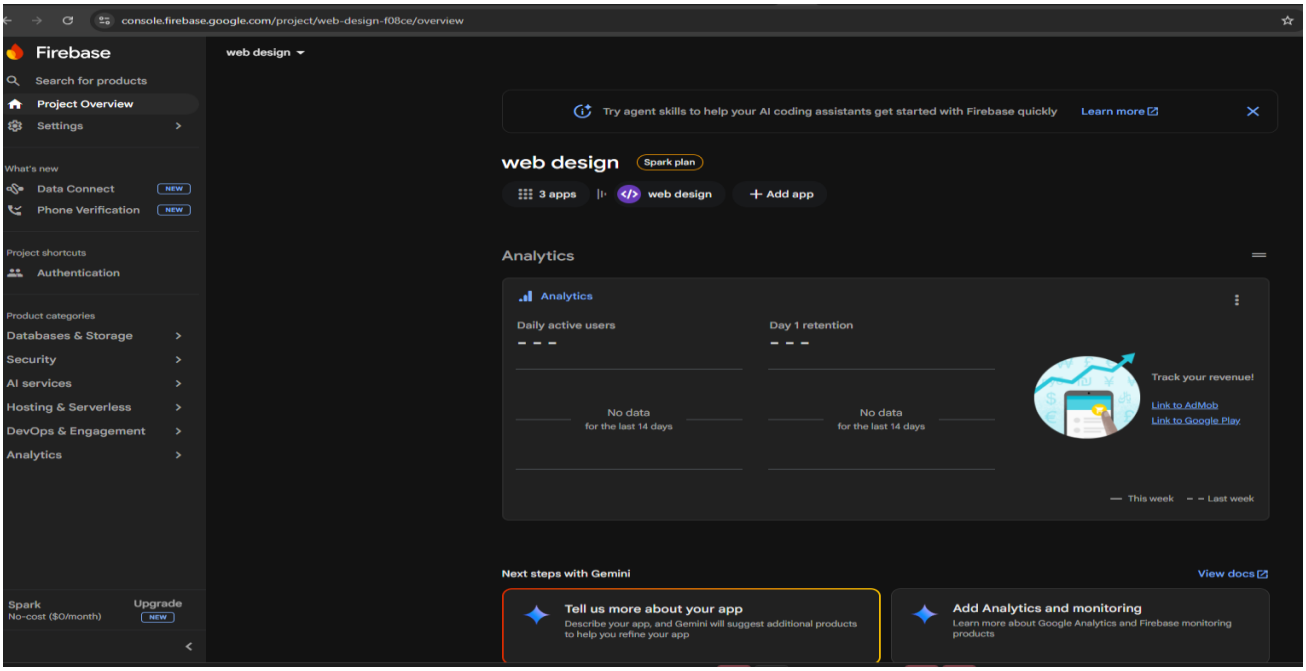


Figure 6 – Firebase Console Cloud Messaging dashboard

1.6 Firebase Console – Compose & Send Notification

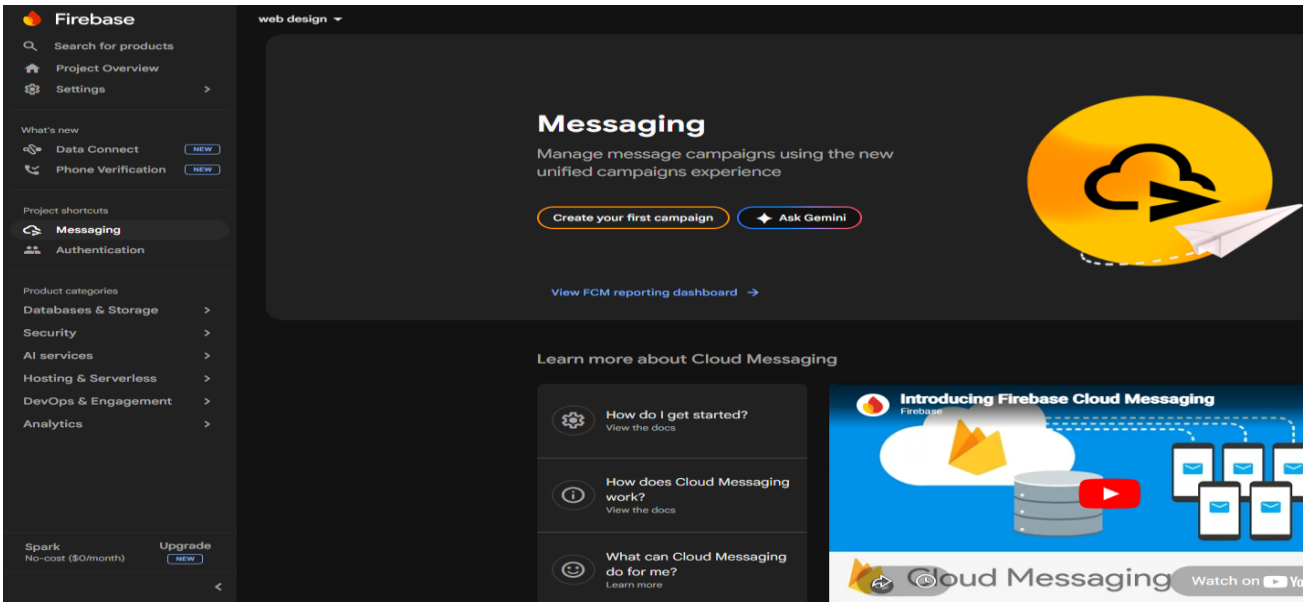


Figure 7 – Firebase Cloud Messaging compose screen

1.7 Firebase Notification Manager

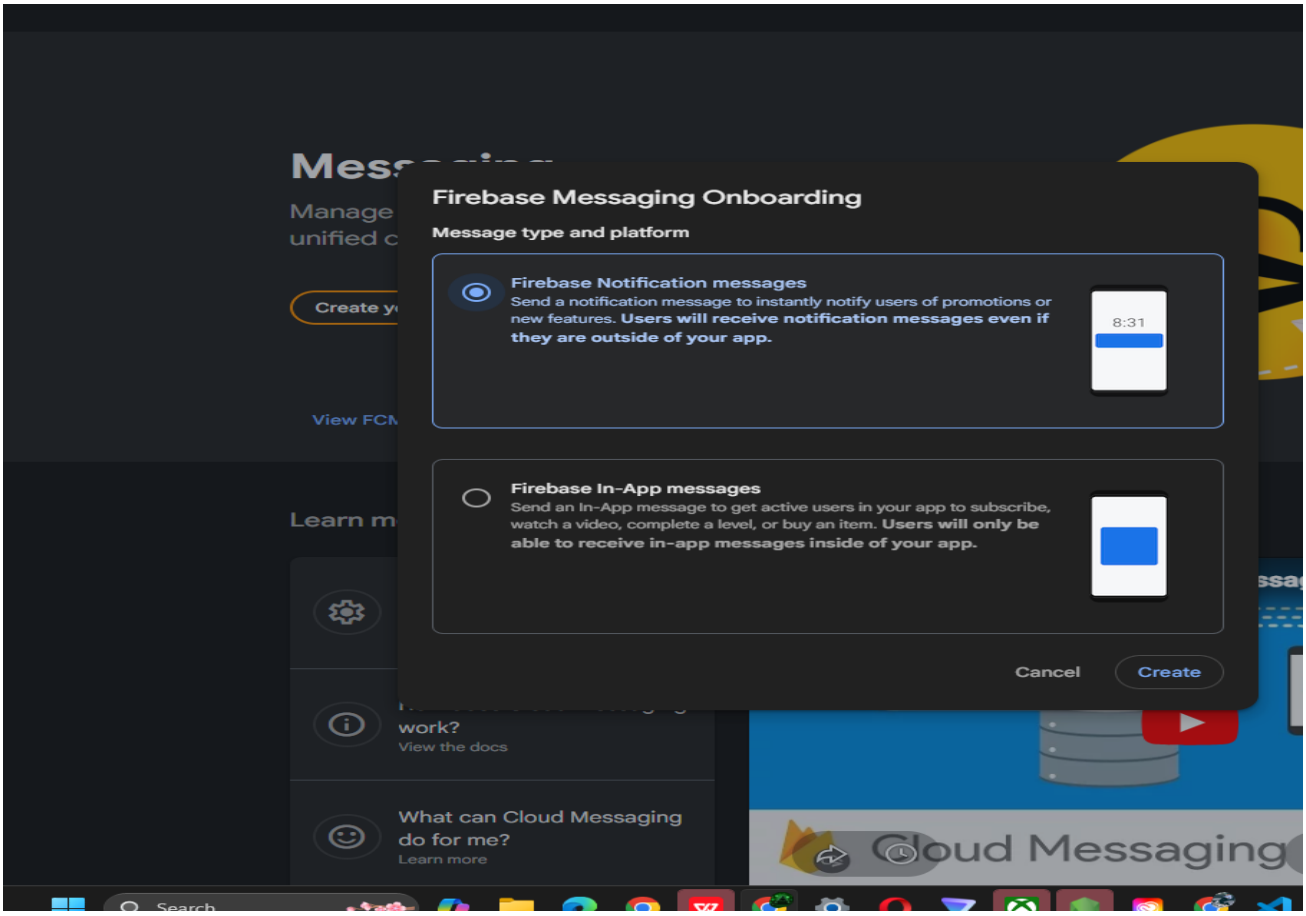
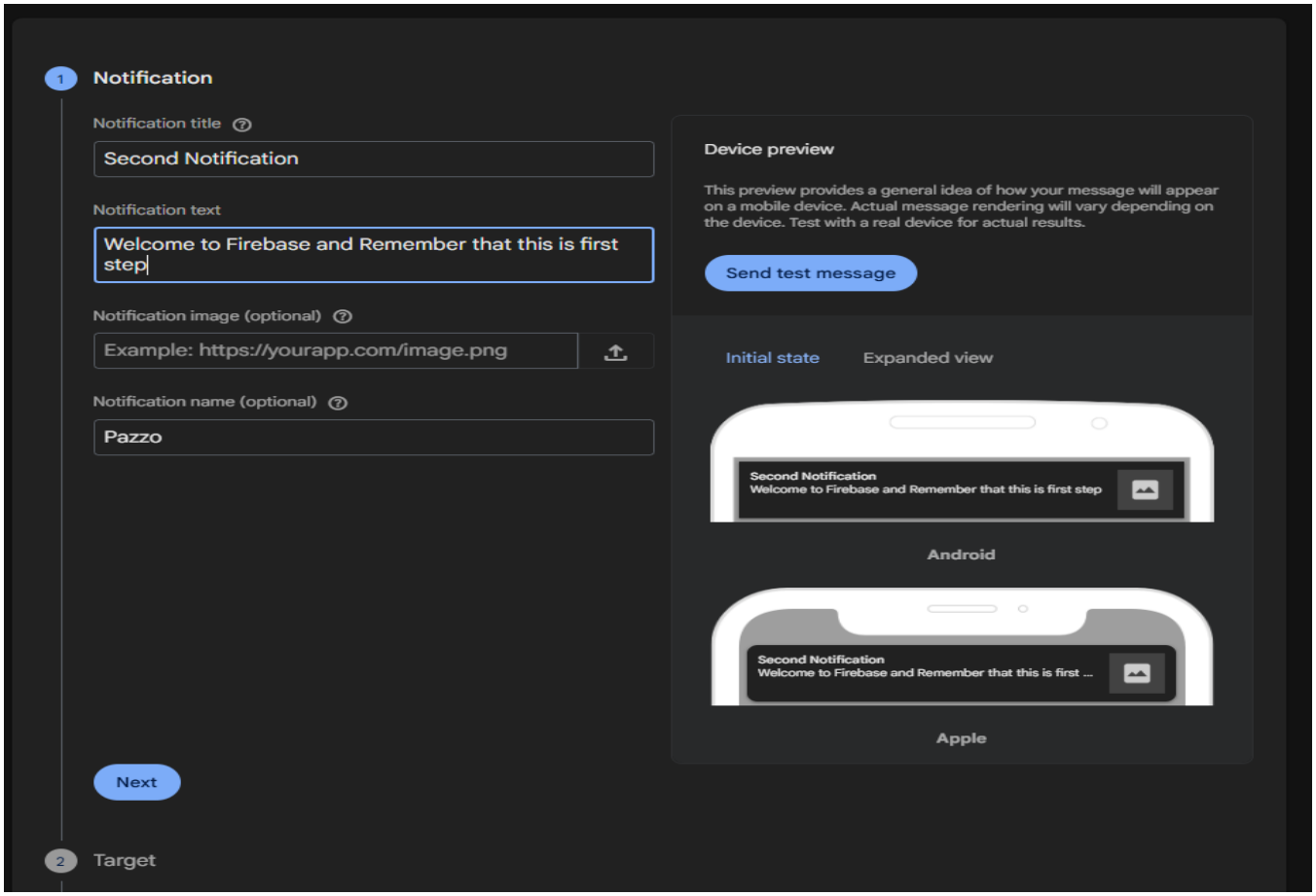


Figure 8 – Firebase Notification Manager with sent campaigns

1.8 Compose your first notification to send



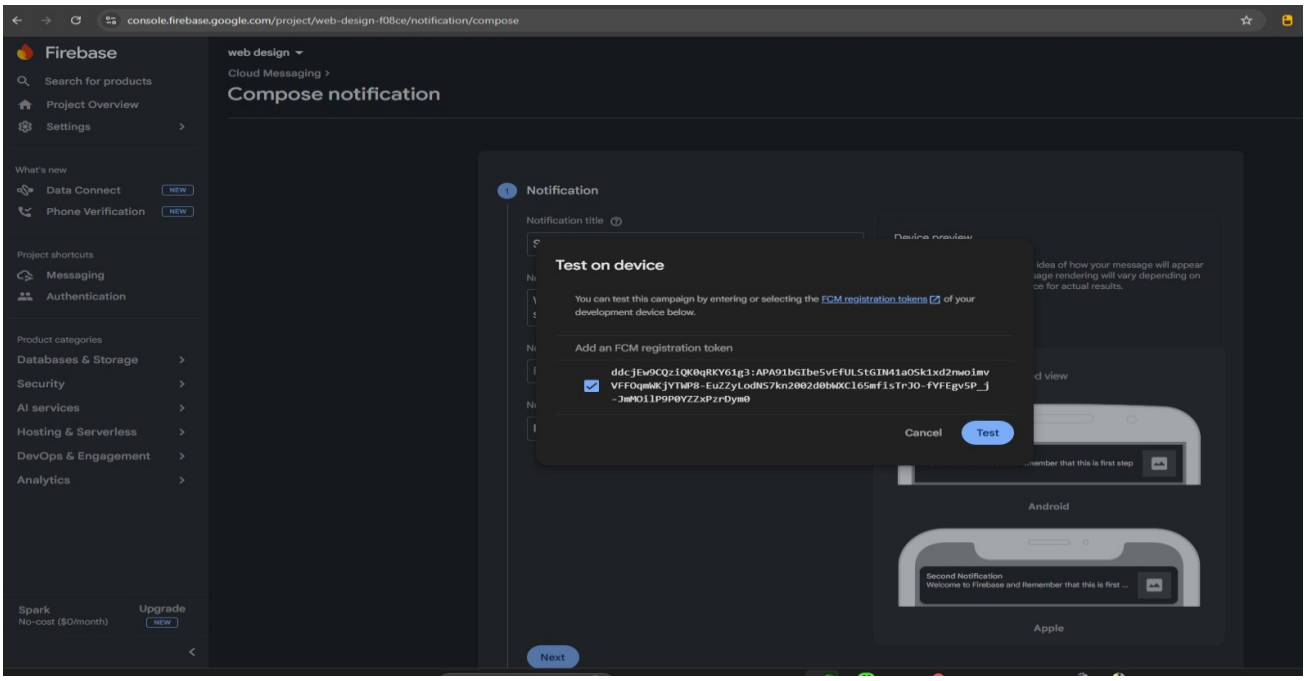


Figure 5 – Device FCM token displayed on screen with Copy button

1.9 Notification Received – App Open (Foreground)

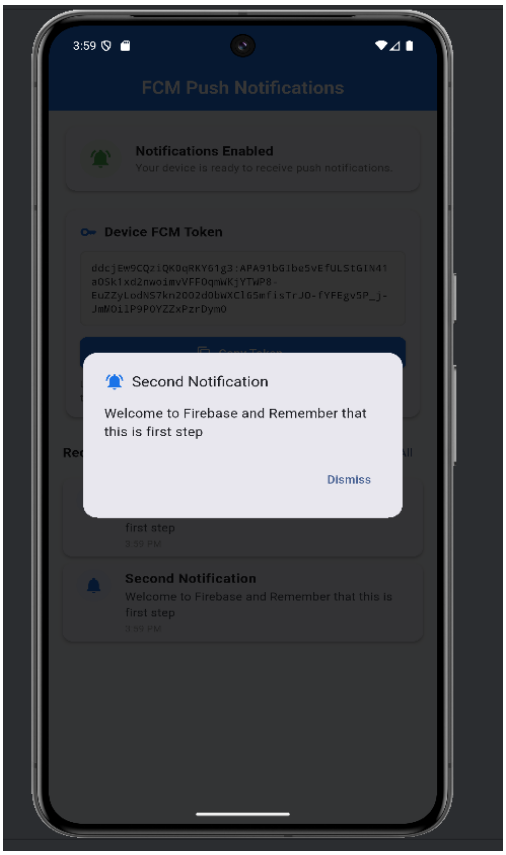


Figure 9 – Notification received while app is open – popup dialog displayed

1.10 Notification Source Detail

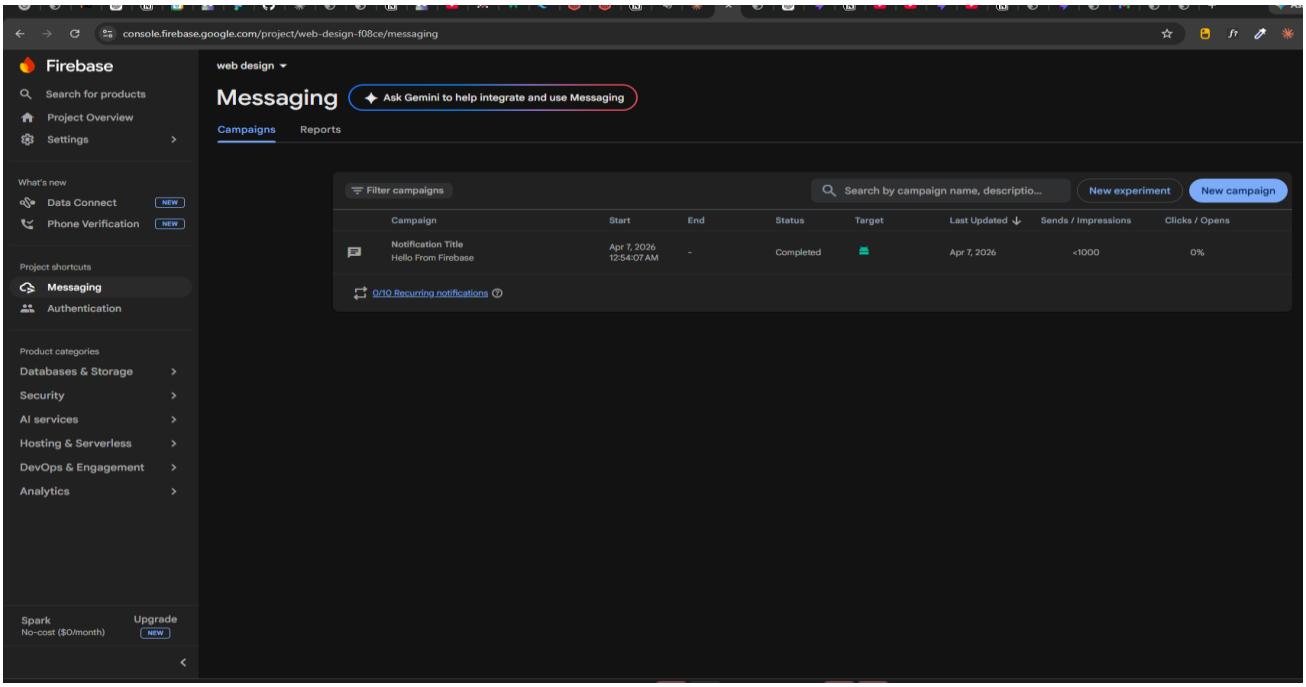


Figure 11 – Notification source and detail view

1.11 Notification in Status Bar – Background State

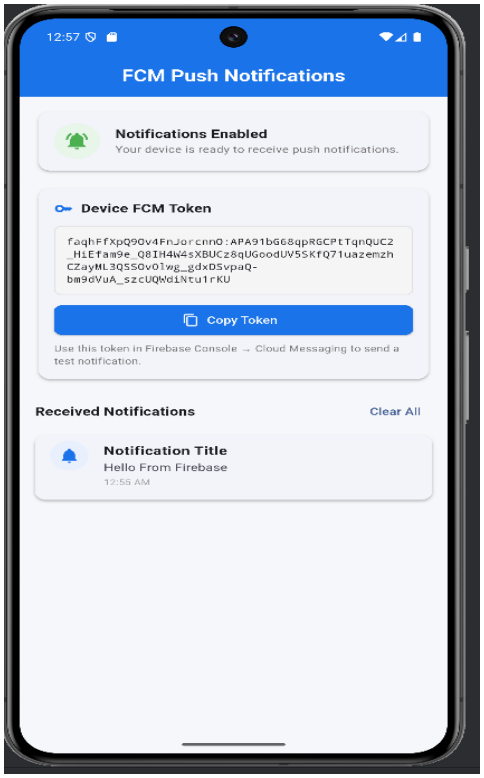


Figure 12 – Notification arriving in status bar A very first notification

1.12 Second Notification Received

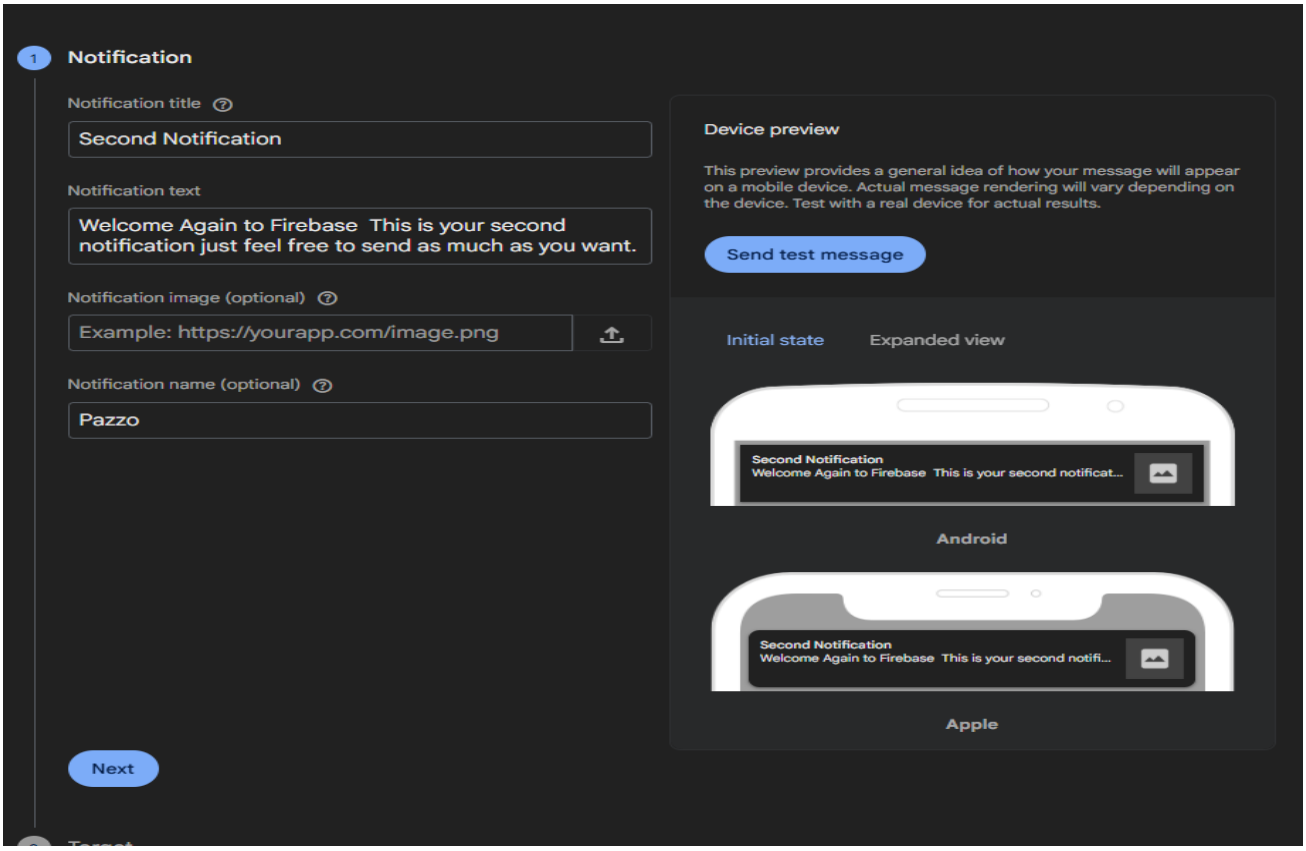


Figure 13 – Second notification received

1.13 Two Notifications in App UI

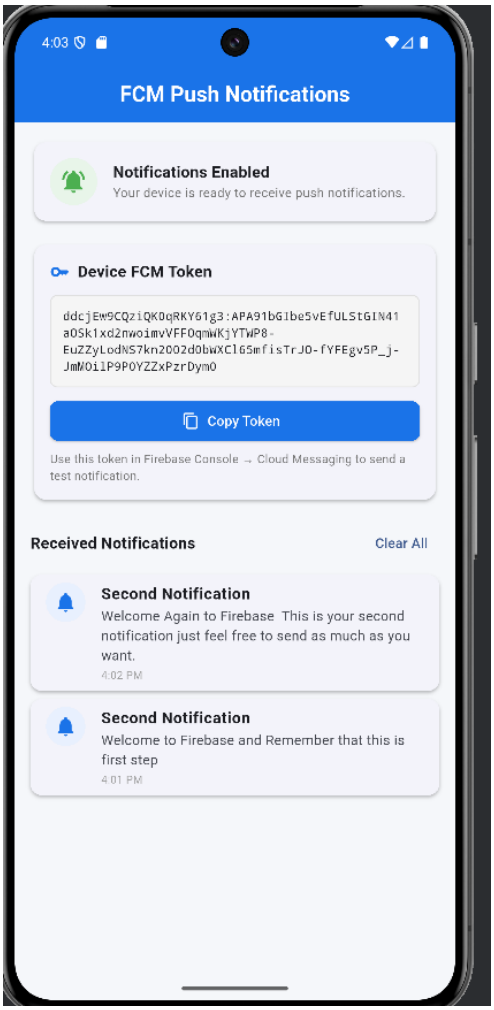


Figure 14 – Two notifications received and displayed in app UI

1.14 Notifications Cleared

After tapping Clear All, the notification list is reset to empty.

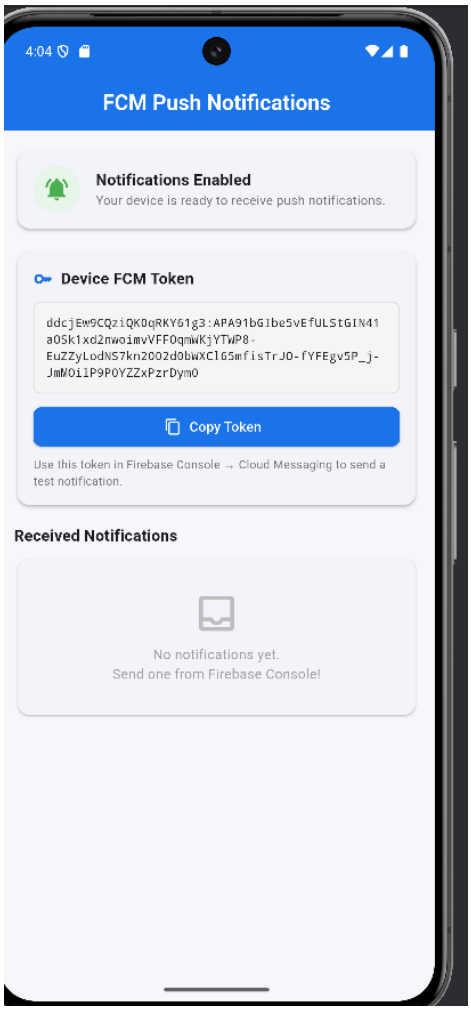
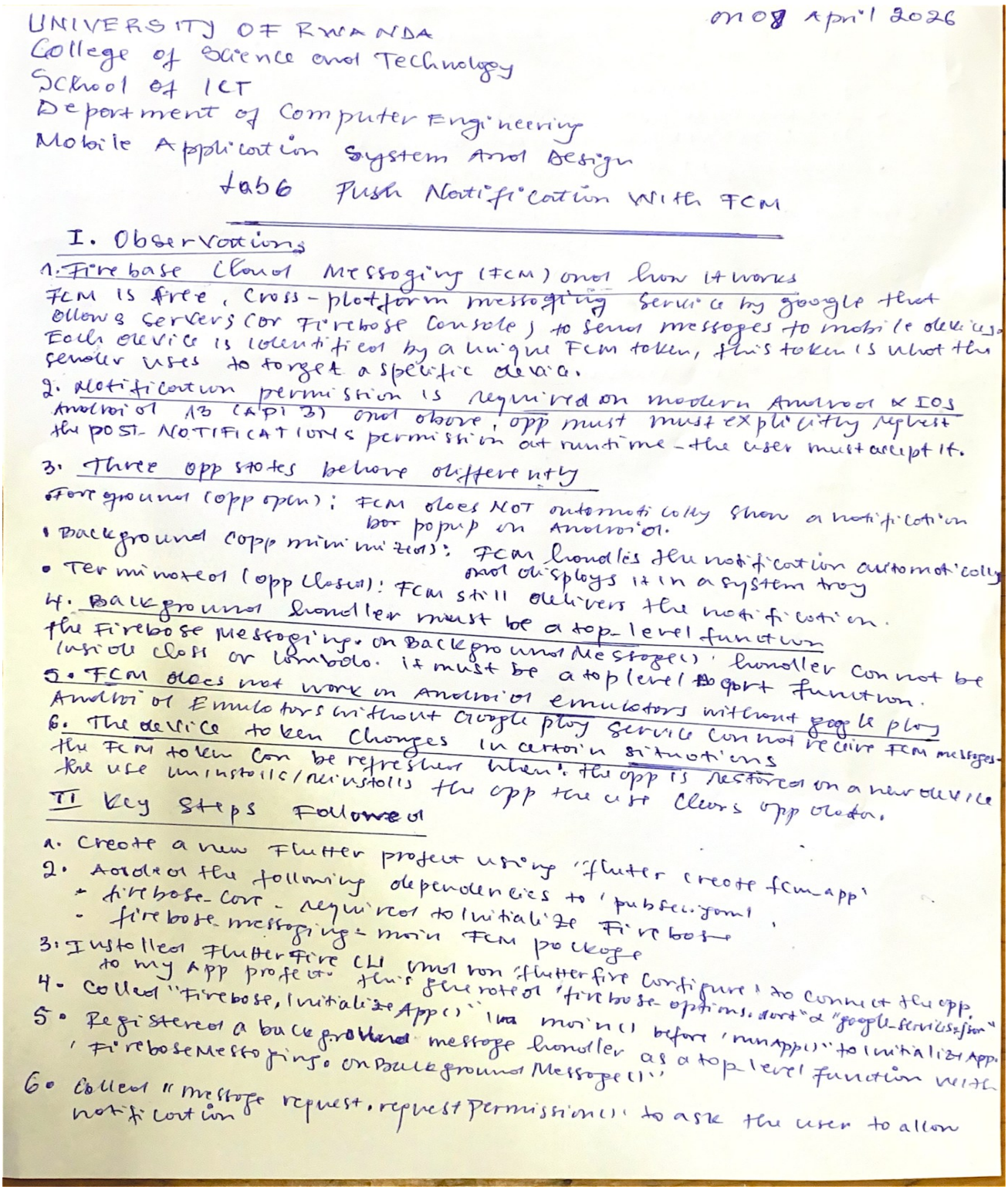


Figure 15 – Notification list cleared

Section 2: Observations & Key Steps (Handwritten)

The following pages contain the original handwritten observations, key steps followed, and difficulties encountered during the lab.



on 08 April 2026

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lab6 Push Notification With FCM

I. Observations

1. Firebase Cloud Messaging (FCM) and how it works
FCM is free, cross-platform messaging service by google that allows servers (or Firebase Console) to send messages to mobile devices. Each device is identified by a unique FCM token, this token is what the sender uses to target a specific device.

2. Notification permission is required on modern Android & IOS
Android 13 (API 33) and above, app must must explicitly request the POST_NOTIFICATIONS permission at runtime - the user must accept it.

3. Three app states behave differently
• Foreground (app open): FCM does NOT automatically show a notification bar popup on Android.
• Background (app minimized): FCM handles the notification automatically and displays it in a system tray
• Terminated (app closed): FCM still delivers the notification.

4. Background handler must be a top-level function
the Firebase Messaging.onBackgroundMessage(): handler cannot be inside class or lambda. it must be a top level function.

5. FCM does not work on Android emulators without google play
Android Emulators without Google play service cannot receive FCM messages.

6. The device token changes in certain situations
the FCM token can be refreshed when: the app is restored on a new device, the user uninstalls/reinstalls the app, the user clears app data.

II Key Steps Followed

1. Create a new Flutter project using 'flutter create fcm_app'
2. Added the following dependencies to 'pubspec.yaml'
   - firebase_core - required to initialize Firebase
   - firebase_messaging - main FCM package
3. Installed FlutterFire CLI and ran 'flutterfire configure' to connect the app to my App project. this generated 'firebase_options.dart' & 'google-services.json'
4. Called "Firebase.initializeApp()" in main() before 'runApp()" to initialize App.
5. Registered a background message handler as a top-level function with 'FirebaseMessaging.onBackgroundMessage()'
6. Called "message.requestPermission()" to ask the user to allow notification

Figure 16 – Handwritten observations and key steps (Page 1)

7. Retrieved the device FCM token with messaging.getToken() and displayed it on screen.

8. Set up three listeners:
   - Firebase Messaging.onMessage → app is open
   - Firebase Messaging.onMessageOpenedApp → App was in background, user tapped notification
   - messaging.getInitialMessage() → app was closed, user tap notification

9. Created an Android notification channel (HIGH_IMPORTANCE_CHANNEL) required on Android 8.0+.

10. Opened Firebase Console → Cloud Messaging → sent a test message using the device token.

11. Received the notification successfully in all three App states.

III Difficulties Encountered

- The emulator did not receive notification easily
- On first launch I forgot to add 'INTERNET' and POST_NOTIFICATIONS permissions to 'AndroidManifest.xml', which caused silent failure.
- The background handler gave an error until I added the @pragma ('vm:entry-point') annotation and made it a top level function.

Figure 17 – Handwritten key steps continued and difficulties encountered (Page 2)

Section 3: Summary

Expected Results Achieved

- Successfully sent a message from Firebase Console
- Received the notification on the phone
- Message content displayed in app notification bar
- Received messages displayed inside the app UI list
- Popup dialog shown when notification received (foreground)
- Notifications received in both open and background states

1. in all three app states